



Lab 07

Design Adder-Subtractor

Name:	ID:	Section:
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7.1 Objectives

After performing this lab, students should be able to:

- Design Verilog modules from given truth table
- Test the functionality of design module through a testbench
- Design Verilog modules from sub-modules using hierarchy
- Design and test Adder-Subtractor module

7.2 Introduction

In Lab 06, 07 and 08, you're going to build a system that is capable of adding and subtracting two 3-bit binary numbers, A and B. The system accepts A, B, and an operator bit, O, which can either be 0 or 1, as input. If the bit O is 0 then the system adds A and B, and when O is 1 it subtracts B from A. In either case, the system generates the appropriate signals for seven-segment display to show the result of addition or subtraction as a hexadecimal digit on Basys-3 board. If the result is negative then the *dp* light of seven segment display is used to indicate that the displayed number is a negative number. The block diagram of the complete system is shown below in Figure 7.1. Before moving to the design, try to convince yourself that system given in Figure 7.1 works as per procedure for addition and subtraction of any 3-bit binary numbers.

1 = Subtraction || Find 2's complement
0 = Addition || Pass through

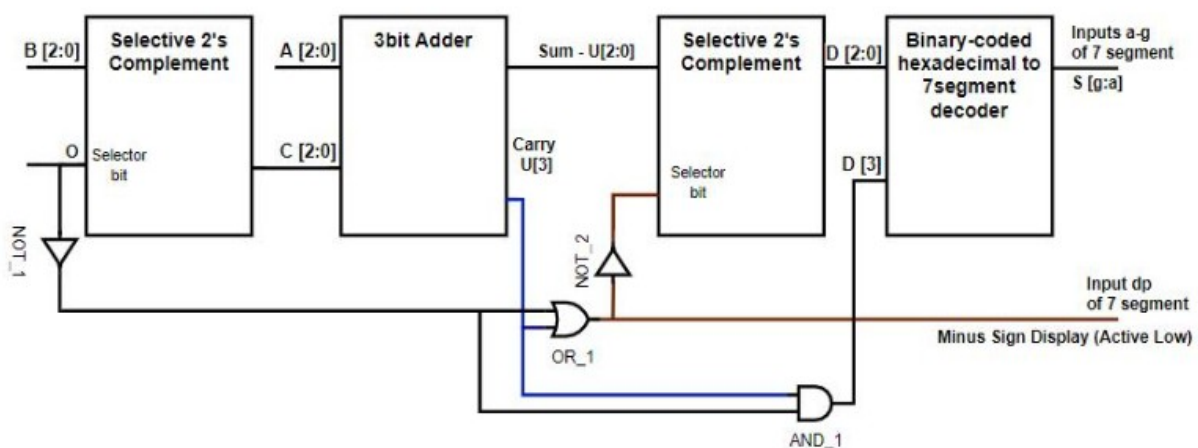


Figure 7.1: Add/Subtract Module



This lab is in continuation with Lab 06, where you have already designed and tested the 3-bit adder module and seven segment decoders. In Lab 07, you will design 2's complement module followed by combining all the designed modules as per above figure. In Lab 08, you will implement the design on BASYS-3 and see it in action. For all labs, beside designing design module, you will design testbench modules as well as you will need to test each design thoroughly before integration. A worksheet is shared for each lab, which you will use to report your results.

7.3 Selective 2's Complement

This circuit accepts four inputs – three bits of the operand B_2, B_1, B_0 and one selection bit O to decide whether 2's complement operation is to be carried out or not, and generates three outputs C_2, C_1, C_0 . If the selection bit is 1 then the circuit computes and outputs 2's complement of the operand, otherwise it simply passes on the operand to output. 2's complement of any number is computed by inverting all its bits and then adding one to the result of inversion.

Task a)

- (i) Complete the truth table similar to Table 7.1 provided in worksheet. As an example, few rows have been filled for you.

Inputs to complement				Outputs of the complement		
O	B ₂	B ₁	B ₀	C ₂	C ₁	C ₀
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	1	0	0
1	0	0	0	0	0	0
1	0	0	1	1	1	1



Table 7.1: Truth table of the Selective 2's Complement module

- (ii) Write Boolean expression for each output in table provided in the worksheet similar to Table 7.2. You should have three expressions i.e., for C_0 through C_2 . Each expression will be in terms of inputs B_0 to B_2 and operator bit O . It is a good idea to simplify these expressions, before we implement them. Use k-map to simplify the expressions. Try simplifying a few expressions by hand or use online Boolean expression simplification tool (<http://www.32x8.com/index.html>) for the rest. Once you learn of K-maps in the coming weeks, you'll be able to directly obtain simplified expressions from truth tables.

$C_2 =$	
$C_1 =$	
$C_0 =$	

Table 7.2: Boolean expression for each output of Selective 2's complement module

- (iii) Write a Verilog module for your selective 2's complement that implements the above expressions in Xilinx Vivado. The module definition is as follows:

```
module complement (  
    input O,  
    input [2:0] B,  
    output [2:0] C );  
    // Write your code here  
    ...  
    ...  
endmodule
```



It is possible to address bits or parts of a vector. For example, $B[2]$ refers to the MSB of variable B.

Submit the following in worksheet for Task a:

- I. Fill out Table 7.1, 7.2.
- II. Appropriately commented code for your design module.
- III. Write a testbench to thoroughly test designed complement module. Test the module for more than 4 different cases.
- IV. Screenshot of Simulation output- make sure to scale properly for visibility of all cases.



7.4 Adder- Subtractor System Integration

Task b)

- (i) Write a Verilog module `addSubtract` that combines all the blocks from Lab 6 and Lab 7 – Task a according to the block diagram in Figure 7.1: Add/Subtract Module. This module's inputs include two 3-bit binary numbers and an operation bit `O`, which is 0/1 for addition/subtraction respectively. The module's outputs are to be connected to the seven-segment display and include the seven control signals for all the segments (a-g) and one bit for the **dp** light of the seven-segment display.
- (ii) Write a testbench to test your adder/subtractor module on following cases:
1. 3+4
 2. 3-4
 3. 7+7
 4. 6-2

Submit the following in worksheet for Task b:

- I. Appropriately commented code for your design module.
- II. Screenshot of RTL Schematic.
- III. Verilog code for testbench.
- IV. Screenshot of output waveform from Simulation - make sure to scale properly for visibility of all cases.

7.5 Exercise

Use Figure 7.1 and work out for cases when `B` is zero and subtracted from `A`. Reflect on the output obtained and suggest ways to fix (if required).



Assessment Rubric Lab 07 Design Adder Subtractor

Name:	ID:	Section:
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Marks Distribution:

		LR2 Code	LR4 Data Collection	LR5 Results	LR10 Analysis
In - Lab	Task a	20	5	10	-
	Task b	20	-	20	-
Exercise		-	-	-	15
Total Marks: 100		/40	/5	/30	/15
CLO Mapping		CLO2	CLO2	CLO2	CLO2

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission and Viva	/10	CLO 4

CLO	Total Points	Points Obtained
2	90	
4	10	
Total	100	

For description of different levels of the mapped rubrics, please refer to the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.





Lab Evaluation Assessment Rubrics

#	Assessment Elements	Level 1: Unsatisfactory	Level 2: Developing	Level 3: Good	Level 4: Exemplary
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/ simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/ simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/ simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/ simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR7	Viva	Response shows a complete lack of understanding of the assigned / completed task	Response shows shallow understanding of the assigned task	Response shows substantial understanding of the assigned task	Response shows complete understanding of the completed task. The student is able to explain all the related concepts.



LR10	Analysis	Results are not interpreted or are analyzed incorrectly.	Analysis of the obtained results is incomplete. Conclusions are not drawn.	Results are analyzed and concluded but are not well explained and justification is not provided with the help of reasoning.	Results are well analyzed supported with reasons. Good comparison is made between the theoretical and experimental results with correct and useful conclusion.
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Lab Evaluation Assessment Rubrics